

2(a)	Volume decreases ; hence w is positive (or small volume change so w is 0) As the ice changes state, heat must have been supplied; hence q is positive. internal energy increases	[1] [1] [1]
(b)	gas expands/increase in volume with work done by gas (against atmosphere) or w is negative the sudden increase leaves little time/no time for thermal energy to enter and leave the gas or $q = 0$ internal energy decreases	[1] [1] [1]

3(a)	molecule(s) rebound from wall of vessel / hits walls/ collide	[1]
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